

National University of Computer and Emerging Sciences
Peshawar Campus

Ideology and constitution of Pakistan
(SS1013)

Date: January 1st 2025

Course Instructor(s)

Gohar Ali Shah. / Muhammad Iqbal. / Tariq Amin

Final-Exam Fall-2024

Total Time (Hrs): 3

Total Marks: 100

Total Questions: 5

Roll No

BSE-1B

Section

Student Signature

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Attempt all the questions.

CLO #:2. Analyze the factors affecting constitutions of Pakistan

Q1: Examine the factors that led to the failure of the 1956 Constitution. [marks= 10]

CLO #1: Describe Pakistan's historical perspective, constitutional phases, contemporary affairs, and future challenges to constitution and democracy in Pakistan.

Q2: Define democracy and analyze the challenges Pakistan faces in the development of democracy? [marks= 10]

CLO #:2. Identifies the importance of good governance, fundamental rights, constitutional and legislative processes in Pakistan.

Q3: The 1973 Constitution of Pakistan guarantees a range of fundamental rights to its citizens. Discuss [marks= 10]

CLO #3: To discuss national institutions, constitutional bodies and constitutional amendments in 1973 constitution of Pakistan.

Q4: Explain the process of electing the President of Pakistan and outline his powers and functions. [marks= 10]

CLO # 3: Discuss national institutions, constitutional bodies and constitutional amendments in 1973 constitution and its impact on the politics of Pakistan.

Q5: How does the 18th Amendment to the Constitution of Pakistan impact provincial autonomy, and why is it significant? [marks= 10]

Functional English (SS1012)

Final- Exam

Date: December 27th 2024

Course Instructor(s)

Ms. Maryam Ali Khan

Mr. Abdul Basit

Total Time (Hrs.)

3

Total Marks:

65

Total Questions:

4

Roll No

PSE-1B

Section

[Signature]
Student Signature

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Attempt all the questions.

CLO 1: Recognize the author's main idea and supporting details. [CLO1= 20 Marks]

The impact of AI on the Jobs market

There is little doubt that new jobs will be required in the AI age. Computer scientists' work will morph into designing programs that specify requirements that will be compiled directly into the code, rather than them writing the code itself, Simmons says. AI is "going to lift the way that software engineers approach their jobs to be much more focused on requirements and specifications and validation and verification than the actual coding practices," he said.

AI won't take away jobs, Simmons insisted, but it will change the nature of jobs. "The way I always describe it is every job consists of a number of tasks and those tasks are going to be replaced by AI and automation, but the jobs will not." They will change the need to do certain things manually, just as word processing eliminated the need for manually typing things, he said. Both Simmons and York say there is already a big demand for people who can do prompt engineering, interacting with LLMs to get the information you want. "That's a job that didn't exist two years ago," said Simmons.

Prompt engineering is "the process of structuring text that can be understood by AI; it's a new way of speaking," said York. "We have abbreviated language because of the Internet, like 'lol'. So we're going to learn to speak in prompts." This makes the human side "more important than ever," York added, because people need to determine where data is coming from and what is missing so they can get accurate information. "The computer can't figure that out," she noted.

Other jobs that will be needed include machine learning engineers, who focus on building, training, and validating models; MLOps (machine learning operations) engineers, who work on testing, deploying, and monitoring models in production; AI testers who can understand the shortcomings of a system's output and fix it, and AI compliance officers who ensure adherence to laws and regulations, said Simmons.

"I firmly believe AI cannot be fully autonomous ... there's always going to be humans and machines working together and the machine is augmenting the human's capabilities," Simmons said. For AI to be successful at augmentation, humans have to determine how systems are designed, the role of the augmenter, how the human's job description changes, and how to create a successful partnership. "Those are questions that need to be asked as new jobs are created," Simmons said.

Q.1 (a): Write the main idea of the text in your own words in one sentence ONLY. [Marks: 04]

Q.1 (b): The article states, "AI cannot be fully autonomous... there's always going to be humans and machines working together." How does the author support this claim? Identify and explain two specific details from the text that emphasize the importance of human involvement in AI-related work. [Marks: 08]

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Q.1 (c) York describes prompt engineering as “a new way of speaking” and emphasizes the need for humans to assess data sources critically. How do these ideas highlight the evolving relationship between humans and AI? Use specific examples from the article to explain your answer. [Marks: 08]

CLO 2- Organize and develop ideas effectively, with logical and well-supported arguments. Develop the ability to read and write for academic purposes through various rhetorical patterns. [CLO2= 20 Marks]

Q.2 The Right to Privacy in the Digital Age: Should Governments Have Access to Personal Data for Security Purposes? [Marks: 05+15=20]

Instructions:

- A. Write an argumentative essay in 5 paragraphs. The essay will not be accepted without brainstorming. You may use any method for brainstorming (e.g., mind mapping, listing pros and cons, etc.).
- B. Argue whether or not government should have the right to access personal data in the name of national security. Take a clear position and provide strong evidence to support your argument. (Follow the typical structure of an argumentative essay as taught in the class).

CLO 3: Make inferences and recognize implied meanings [CLO3= 10 Marks]

In an era dominated by instant communication, where messages are exchanged with a mere tap on a screen, many individuals still hold onto the tradition of handwritten letters. They argue that despite the convenience and speed of digital messages, there is something profoundly meaningful about a handwritten note. For them, the act of writing by hand is a deliberate and intimate gesture, imbued with emotion and sincerity. Unlike the fleeting nature of text messages or emails, which can often be composed hurriedly, a handwritten letter is seen as a tangible reflection of thoughtfulness and personal investment. Each word is carefully chosen, each sentence crafted with time and attention. In contrast, digital communication—though efficient—can sometimes feel impersonal or even superficial, leaving little room for the emotional weight that a handwritten letter can carry. For these individuals, the act of writing and receiving a handwritten letter represents more than just communication; it signifies a deeper connection, a lasting memory, and a reminder that in a world increasingly defined by immediacy, some things are worth taking time for.

Q3: (a) What idea can be inferred from the paragraph above? (Maximum 20 words)

Q3: (b) Justify the inferences you make from the paragraph above. (Maximum 50 words)

CLO 4: Identify literary devices and analyze the texts [CLO4= 15 Marks]

Q. 4: Provide one example each of verbal, situational, and dramatic irony, using clear headings for each type. Use examples from your experience, a book, a movie, or current events. Limit each example to a single paragraph to demonstrate conceptual clarity instead of providing definitions. [Marks: 05+05+05=15]

Calculus & Analytical Geometry (MT1003)

Final-Exam

Date: 24 Dec 2024

Total Time (Hrs): 3

Course Instructors

Total Marks: 90

Mr. Osama Sohrab / Dr. Shams Ul Arifeen

Total Questions: 9

Roll No: _____

Section: BSE-1B

Student Signature:

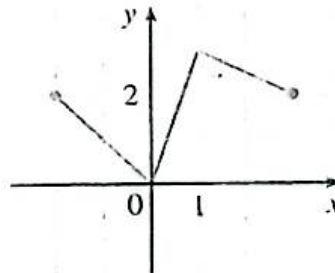
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Attempt all the questions. Each question carries ten marks.

CLO # 1: Apply knowledge of mathematics, science and appropriate knowledge domain to mathematically model and conceptualize defined problems.

Q1. The graph of f is given. Draw the graphs of the following functions:

(a) $y = f(x + 1)$ (b) $y = -f(x)$ (c) $y = \frac{1}{2}f(x)$



Q2. Use chain rule to compute $\left. \frac{dy}{du} \right|_{u=1}$, where

$$y = s^2 e^s, \quad s = 2^{t+1}, \quad t = \sin u$$

Q3. Approximate $\sigma(0.1)$ using linearization at $x = 0$, where

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Q4. Use l'Hospital's Rule to evaluate the limit

$$\lim_{x \rightarrow 0} \frac{1}{x^2} \int_0^x \frac{2t}{\sqrt{t^3 + 1}} dt$$

CLO # 2: Use fundamental principles of mathematics and relevant domains to identify, analyze and solve problems in order to reach substantiated conclusions.

Q5. Evaluate the following integrals:

(a) $\int \cos^3(\pi\theta) \sin^4(\pi\theta) d\theta$

(b) $\int \frac{dx}{x^2 - 7x + 10}$

Q6. In a certain city, the temperature (in °C) t hours after 9 am was modeled by the function

$$T(t) = 10 + 4 \sin\left(\frac{\pi t}{12}\right)$$

Find the average temperature during the period from 9 am to 9 pm.

Q7. Find the area of the region bounded by the graphs of $x = 3 - y^2$ and $x = y + 1$.

Q8. On what interval is the curve

$$f(x) = \int_0^x (1 - t^2) e^t dt$$

increasing?

Q9. Evaluate $\int_{-\infty}^{\infty} f(x) dx$, if

$$f(x) = \begin{cases} xe^{-x^2}, & x < 0 \\ \frac{1}{\sqrt{1-x}}, & 0 \leq x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

THE END

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Applied Physics (NS1001)

Date: Dec 19th, 2024

Course Instructor(s)

Engr. Asif khan/Dr. Suleman Mir/

Dr. Maqsood M. Khan

Final Exam

Total Time (Hrs): 3

Total Marks: 50

Total Questions: 4

Roll No

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Attempt all the questions.

CLO 1: Comprehend the fundamental laws of physics relevant to computer science and software engineering

Question 1: An astronaut has left an orbiting spacecraft to test a new personal maneuvering unit. As she moves along a straight line, her partner on the spacecraft measures her velocity every 2.0 s, starting at time $t = 1.0$ s. The records are listed in table 1.

Time (sec)	Vx (m/sec)	Time (sec)	Vx (m/sec)
1	0.8	9	-0.4
3	1.2	11	-1
5	1.6	13	-1.6
7	1.2	15	-0.8

Table 1

(A) Plot the velocity for the astronaut. [1.67 M]

(B) Calculate and plot average acceleration, and state whether the speed of the astronaut increases or decreases over each of these 2.0-s time intervals: (i) 1 to 3 secs (ii) 5 to 7 secs (iii) 9 to 11 secs (iv) 13 to 15 secs [5M]

CLO 2: Apply knowledge of basic physical laws to solve various problems of applied nature

Question 2:

(A) Sound waves are longitudinal waves in air. The speed of sound depends on temperature; at 20°C it is 344 m/sec. Calculate the wavelength of a sound wave in air at 20°C, if the frequency is 262 Hz. [4.67 M]

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(B) A moving pulse is represented by the wave function as,

$$y(x, t) = \frac{4}{(x + 3.0t)^2 + 1}$$

Figure 1 shows the pulse represented by this wave function at $t = 0$, where x and y are measured in centimeters and t is measured in seconds. Find and plot expressions for the wave function at $t = 0$, $t = 1.0$ s, and $t = 2.0$. [6M]

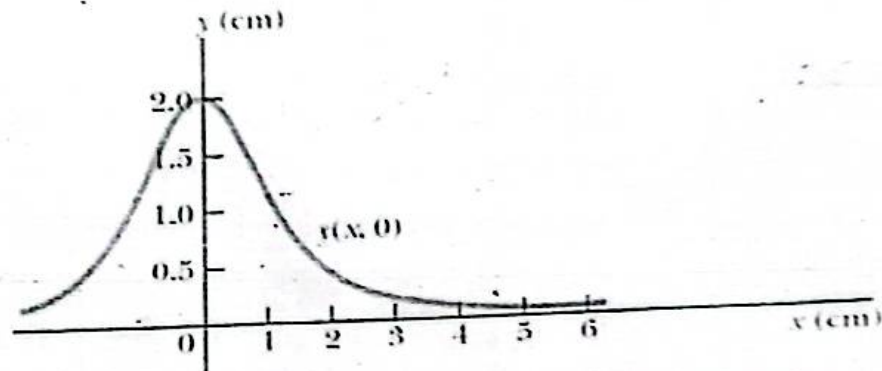


Figure-1

(C) Two wave's functions y_1 and y_2 are represented as individual wave functions:

$$y_1 = 2 \sin(5.0x - 4.0t) \quad \text{and} \quad y_2 = 2 \sin(5.0x + 4.0t)$$

where x and y are measured in centimeters.

- (i) Find the amplitude of the simple harmonic motion of the element of the medium located at $x = 2.3$ cm.
- (ii) Find the positions of the nodes and antinodes if one end of the string is at $x = 0$. [6M]

CLO 3: Analyze different physical problems using the laws of physics from different areas like circuits and electronics.

Question 3:

(A) Discuss the following in not more than three lines. Your discussion must be comparative.

- | | |
|---|------|
| (i) Voltage, potential difference and electromotive force | [2M] |
| (ii) Electric field and magnetic field | [2M] |
| (iii) Ampere's and Coulomb's laws | [2M] |

(B) Solve the circuit shown in figure 2, to compute the equivalent resistance across terminals a & b. [10M]

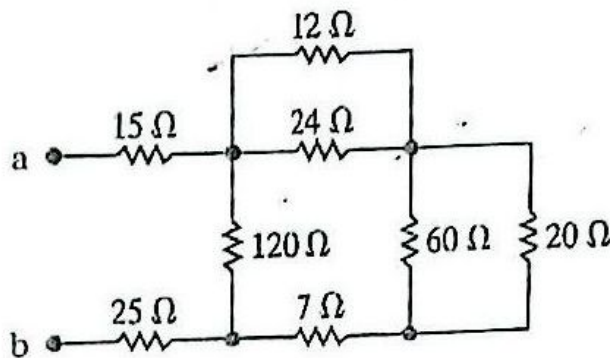


Figure 2

Question 4:

- (A) Briefly explain (not more than three lines) how electromagnetic wave can be generated. [2.66M]
- (B) A resistor "R" is connected across a source having voltage "V" and current "I" with an internal resistance "r". Calculate the following:
- (i) If $V = 1\text{kV}$, $I = 5\text{mA}$ & $r = 2\Omega$: what will be the exact E.M.F of the source and what will be the value of R? [2M]
 - (ii) For part (i) show that power delivered is equal to power dissipated in "R". [2M]
 - (iii) What will be the exact power of the source if "R" is not connected? [2M]
 - (iv) If $R = 300\Omega$ and the power dissipated by R is 480mW what will be I & V? [2M]

The End

Programming

Fundamentals (CS1002)

Date: 16th December, 2024

Course Instructor(s)

Samin Ahmed, Sara Rehmat,
Usman Wajid

Roll No

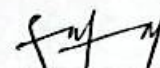
BSE-1
Section

Sessional-II Exam

Total Time (Hrs): 3

Total Marks: 100

Total Questions: 9


Student Signature

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Attempt all the questions.

CLO 1: Understand Basic problem solving steps and logic constructs.

Q1: Give short answers to the following questions:

[marks: 2.5*4=10]

- What is the difference between a function call by value and a function call by reference?
- What is the difference between an integer variable, an integer pointer and an integer pointer-to-pointer variable?
- Write the starting memory address of the index in the following integer 2D array in which the value is 100.
1. Assume that the base address of the following integer 2D array is 100.

- What is file handling and how it is benefited over the hardcoding? Also write the data types (stream objects) for creating variables for opening files to perform input and output operations

Q2: Consider the following two pieces of C++ code and provide the output of the program. Assume all necessary header files to be included.

[marks: 5*2=10]

a.

```
int main() {  
    int num[5] = {400, 50, 200, 300, 800};  
    int* ptr1 = num;  
    int* ptr3 = (int*) (((char*)ptr1) + 4);  
    //int* ptr3 = static_cast<int*> ((static_cast<char*>(ptr1)) + 4);  
    int* ptr2 = ptr1 + 2;  
    *(ptr2 - 1) = 99;  
  
    cout << "ptr1: " << *ptr1 << endl;  
    cout << "ptr2: " << *ptr2 << endl;  
}
```

```
cout << "ptr3: " << *ptr3 << endl;
cout << "ptr3+1: " << *(ptr3+1) << endl;
cout << "ptr2+1: " << *(ptr2+1) << endl;
return 0;
}
```

b.

```
bool func(char* p, int n) {
    char* q = p;
    char* r = p + n - 1;
```

```
    while (q < r) {
        if (*q != *r) {
            return false;
        }
        q++;
        r--;
    }
    return true;
}
```

```
int main() {
    char data[] = "radar";
    int n = strlen(data);
    bool result1 = func(data, n);
    strcat(data, "s");
    bool result2 = func(data, n+1);
    result1? (cout << "Yes" << endl) : (cout << "No" << endl);
    result2? (cout << "Yes" << endl) : (cout << "No" << endl);
    return 0;
}
```

Q3: Highlight logical or syntactical error(s), if any, in the following pieces of code. Rewrite the code after removing the errors. Read the code from left column to right column. [marks: 5*2=10]

a.

```
#include <iostream>
using namespace std;
int main()
{
    int* arr = new int[5];
    int* p;
    for(int i=0; i<5; i++)
        cin >> arr[i];
    delete arr;
    p = arr;
    for(int i=0; i<5; i++)
        p[i]=1;
}
```

return 0;

}

b.

```
#include <iostream>
using namespace std;
int x = 10;

int main()
{
    int x = 5;
    for(int i = x; i<10; i++)
    {
}
```

```

int x = 15;
cout << x << endl;
}

cout << x << endl;
cout << i << endl;
cout << calculate_area('a', 6) <<
endl;

return 0;
}

int calculate_area(int l, int w)
{
    int area = l*w;
    cout << area << endl;
}
    
```

CLO 2: Apply basic programming concepts.

Q4: Write a program in C/C++ that

[marks: 10]

- Take three integers from the user
- Passes them to a function named `calc_stats()` that calculates *mean* and *range* (difference between maximum and minimum) of these integers
- Returns the calculated mean and range to the `main()` function without using arrays
- Displays the mean and range in the `main()` function

Q5: Write a program in C/C++ that

[marks: 10]

- Initializes an integer array of size 10 in run time
- Declares another integer named `test_int` and takes its value from the user
- Counts the number of occurrences of `test_int` in the array using pointer notation.
- Display the number of occurrences of `test_int`

(Note: Solution without the usage of Pointer notation will be given 0 credits)

Q6: Write a program in C/C++ that

[marks: 10]

- Takes two strings from the user
- Sends the first string to a user defined function named `rev_string` that reverse's the strings.
- Sends the reversed first string and the second string to a user defined function named `cat_string` that concatenates both the strings into a single string
- Displays the resultant string in the `main()` function.

(Note: Usage of built-in string functions will result in 0 credits)

CLO 3: Design and implement algorithms in C/C++ to solve real world problem.

Q7: Define overloaded functions for merging two integer arrays into a single sorted array and two strings into a single string.

[marks: 10]

Q8: You are tasked with creating a program in C++ that can encrypt text files. The encryption will involve shifting each character in the file by a specified number of positions.

[marks: 15]

- Read a file and encrypt its content using the method described in following steps:
 - Shift each alphabetic character (both uppercase and lowercase) by a user-provided integer key.
 - Validate user input for encryption keys to ensure they are integers.
 - Non-alphabetic characters (spaces, punctuation, numbers, etc.) should remain unchanged.

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- Write the encrypted content to a new file.
- Handle scenarios where the file cannot be opened (e.g., file does not exist).

Q9: Write a program that processes a 4x4 grayscale image by converting it to a binary image, resizing it and flattening it. The conversion is performed using a certain value called threshold value. A 4X4 grayscale image has values for pixel intensities in range 0 to 255 inclusive. A binary image has each pixel intensity value either as 0 or 1. [marks: 15]

- The program must take pixel values for the grayscale image and the threshold value from the user.
- The program must ensure that all pixel values as well as the threshold value are between 0 and 255 (inclusive). If an invalid value is entered either for pixel intensity or threshold, the user is prompted again till a valid value is entered.
- The conversion is performed by calling a function that is passed the grayscale image and threshold value from the main function. The function should replace each pixel value with 1 if the intensity is greater than or equal to the threshold value and with 0 otherwise.
- The binary image is then resized to a 2x2 image by calling a function. The function computes each pixel in the resized image as the average intensity of a corresponding 2x2 block from the original image.
- The resized image is then flattened by calling a function that converts the image to a one dimensional array.
- The program must display the pixel values before and after each processing of the image. Please keep in mind that input from keyboard and output to console should only be done in the main function.

**National University of Computer & Emerging Sciences
Peshawar Campus**



Student Name: _____

Roll No: _____

Program: SE

Examination: Final

Semester: Fall 2024

Total Marks/Weightage: 50

Time Allowed: 03 hours

Instructor: Iqra Rehman

Course: Programming Fundamentals

Date: 5th December 2024

NOTE:

- Attempt all questions.
- Don't rush, carefully understand the problem and then go for solution.
- Understanding the question is the part of Exam.
- An individual may be assigned a straight-forward 0 if the submitted assessed task is copied/cheated from another individual/internet/found using internet.
- Create a folder named as Roll number, name and section, copy all the files (Q1.cpp, Q2.cpp,), in that folder and submit the folder.
- Program should also have proper comments.

Question#1

You are tasked with developing a C++ program that identifies the longest word in a user-input sentence.

Requirements:

1. The program should be able to read a sentence from the user.
2. It should identify and display the longest word in the sentence.
3. If multiple words have the same maximum length, the program should display the first one.
4. The program must use functions to separate logic:
 1. One function for reading the sentence and displaying the result.
 2. Another function for identifying the longest word.

5. The solution should not use any built-in string manipulation functions.
Weightage: [10]

Question#2

Imagine you have a list of numbers that are supposed to form a continuous sequence, but some numbers are missing. For example, you might have a sequence like 1, 2, 4, 5 where the number 3 is missing. Your task is to create a C++ program that helps identify these missing numbers. The user will input a sequence of numbers, and the program will check for any numbers within this range that are not present in the input and display them as missing.

Sample output:

Enter a sequence of numbers: 1 2 4 5 7

Missing numbers: 3 6

Weightage: [10]

Question#3

The Game of Life is a simulation created by a mathematician named John Horton Conway in 1970. It takes place on a grid (like a checkerboard) made up of rows and columns of cells. Each cell can either be "alive" (represented by 1) or "dead" (represented by 0). Each cell interacts with the eight cells around it (above, below, to the sides, and diagonally). Depending on the number of "alive" neighbors, the state of each cell changes according to these simple rules:

1. **Underpopulation:** If a live cell has fewer than two live neighbors, it dies.
2. **Survival:** If a live cell has two or three live neighbors, it stays alive.
3. **Overpopulation:** If a live cell has more than three live neighbors, it dies.
4. **Reproduction:** If a dead cell has exactly three live neighbors, it becomes alive.

To update the grid, you apply these rules to every cell at the same time, creating the "next state" of the board. This means cells "die" or "come to life" all at once, based on the current state of their neighbors.

Your task is to take the current state of the board (4X3 matrix) from user, and pass it to function named NextState() and apply these rules, and update it to show what it will look like in the next round and display the next state.

$0,0$ 0	$0,1$ 1	$0,2$ 0
$1,0$ 0	$1,1$ 0	$1,2$ 1
$2,0$ 1	$2,1$ 1	$2,2$ 1
$3,0$ 0	$3,1$ 0	$3,2$ 0



$k-1, l-1$ 0	$k-1, l$ 0	$k-1, l+1$ 0
$k, l-1$ 1	k, l 0	$k, l+1$ 1
$k+1, l-1$ 0	$k+1, l$ 1	$k+1, l+1$ 1
0	1	0

Sample output:

Input:

```
board = [[0,1,0],
          [0,0,1],
          [1,1,1],
          [0,0,0]]
```

Output:

```
[[0,0,0],
 [1,0,1],
 [0,1,1],
```

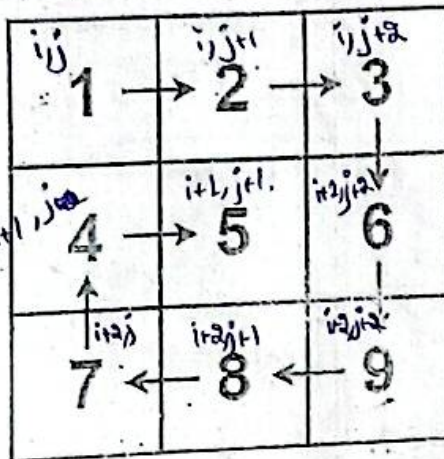
[0,1,0]]

Weightage: [15]

Question#4

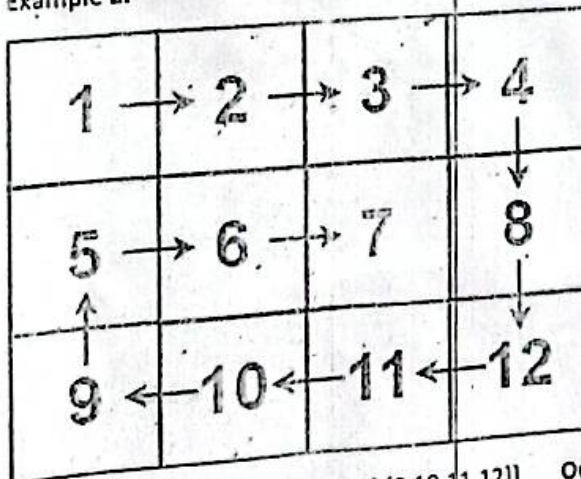
Given an $m \times n$ matrix, return all elements of the matrix in spiral order.

Example 1:



Input: matrix = [[1,2,3],[4,5,6],[7,8,9]] Output: [1,2,3,6,9,8,7,4,5]

Example 2:



Input: matrix = [[1,2,3,4],[5,6,7,8],[9,10,11,12]] Output: [1,2,3,4,8,12,11,10,9,5,6,7]

Weightage: [15]

ICT Lab BSE-1B(CL1000)

Final Exam

Date: Dec 03 2024

Course Instructor

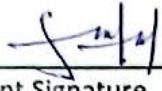
Mr. Muhammad Qasim

Total Time: 1 hour, 30 min

Total Marks: 50

Roll No

BSE-1B
Section


Student Signature

Attempt all the questions.

Create a folder, named as: RollNo_yourname_section. Include all the solution files in this folder and the name of each file should have format as: RollNo_QuestionNo.

Q1: Write short answers for the following questions.

- What is a CPU? (2 marks)
- What is application software and system software? Give examples. (2 marks)
- What is BIOS? Give examples. (2 marks)
- You want to create multiple LAN's, they should communicate with each other, which devices should you use to make these LAN's functional? (2 marks)
- What these cmd commands does? (2 marks)
Dir *.exe
Dir *.*
TIME
Cd ..

Q2: Create a document in MS word having format same as the image given in the figure 01. Use different fonts for heading, text etc. Students can generate random text using =rand() function and can insert text in the document. Insert an image of your own choice but the structure of the

document should be same as the image given below. (10 marks)

Yahya reiterates pledge to defend Pakistan *Crisis has forged a stronger nation*

RAWALPINDI, Aug 13: The President, Gen A.M. Yahya Khan, has called upon the nation to stand united to defend the sovereignty and integrity of the country against internal subversion and threat of external aggression.

In an Independence Day



message to the nation, the President reiterated his pledge to preserve and defend the country which he described "a sacred trust." "The creation of Pakistan itself is a tribute to the

relentless and united struggle of its people, under our great leader, the Quaid-i-Azam", he added.

He said the nation is celebrating the birthday of the homeland in the shadow of grave events. The unity and integrity of Pakistan had been challenged by internal subversion and threat of external aggression.

The President said "by the grace of Allah, the united will of the people has averted disaster, and reaffirmed its unshakable faith in the sovereignty and integrity of Pakistan."

TRIBUTE TO PEOPLE

The President said: "we celebrate the birthday of our homeland today in the shadow of grave events. The unity and integrity of Pakistan have been challenged by internal subversion and threat of external aggression."

"By the grace of Allah, the united will of the people has averted disaster, and reaffirmed its unshakable faith in the sovereignty and integrity of Pakistan, it is my conviction that the crisis has forged a stronger and prouder nation, determined to move ahead, and thwart the designs of its enemies."

"The creation of Pakistan

itself is a tribute to the relentless and united struggle of its people, under our great leader, the Quaid-i-Azam. Their resources of mind and spirit overcame the hostile forces of colonial rule and Hindu domination.

"These forces are still not reconciled to the existence of Pakistan. I do not wish to minimise the present danger to our integrity. The imperatives of the situation demand that we should close our ranks and stand united. Pakistan is a sacred trust. On this solemn occasion, I reiterate my pledge to preserve and defend this trust."

"The need of the hour is to revive the same spirit and dedication which brought us the supreme gift of independence, and saw emerge triumphant over every challenge in the past. History is witness to the courage and faith displayed by the people of Pakistan. Let us stand firm on this bedrock of national unity in the years that lie ahead. Let every one

(EDITORIAL ON PAGE 9)

of us wherever he may be do his duty faithfully and courageously, repose complete faith in Almighty Allah, and move forward to the fulfillment of our great destiny. Pakistan Painsdabad."—APP.

Figure 01

Q3.

a. Fill the following table and also write the steps you used to find these conversions. (6 marks)

Decimal	Binary	Hexadecimal	Octal
(48) ₁₀	(10000) ₂	(30) ₁₆	(60) ₈

b. $(1100)_2 + (100)_2 = (?)_{16}$

(4 marks)

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Q4. You have these two sheets (sheet1, sheet2) in an excel workbook. You need to fill the Grade column in sheet1. Write VLOOKUP function to fill this column in sheet1 (10 marks)

	A	B	C	D	E	F			
1	StudentID	Name	Course	Grade	1	StudentID	Name	Course	Grade
2	ST101	John Smith	Math		2	ST101	John Smith	Math	85
3	ST102	Emma Davis	Math		3	ST102	Emma Davis	Math	92
4	ST103	Michael Lee	Physics		4	ST103	Michael Lee	Physics	78
5	ST104	Sarah Wilson	Chemistry		5	ST104	Sarah Wilson	Chemistry	95
6	ST105	James Brown	Physics		6	ST105	James Brown	Physics	88
7	ST106	Lisa Chen	Math		7	ST106	Lisa Chen	Math	91
8	ST107	David Miller	Chemistry		8	ST107	David Miller	Chemistry	62
9	ST108	Anna Taylor	Physics		9	ST108	Anna Taylor	Physics	89
10	ST109	Ryan Garcia	Math		10	ST109	Ryan Garcia	Math	77
11	ST110	Sophie Martin	Chemistry		11	ST110	Sophie Martin	Chemistry	94
12					12				
13					13				
14					14				
15					15				
16					16				
17					17				
18					18				
19					19				
20					20				
21					21				
22					22				
23					23				
24					24				
25					25				
26					26				
27					27				
28					28				

Sheet1

Sheet2

+

Sheet1

Sheet2

+

Q5. Write html code and external css to design the following form (10 marks)

Sample Form

Name:

Enter your name

Email:

Enter your email

Choose your interests:

☐ Coding

☐ Sports

☐ Music

Select your country:

-Select-

Submit